

A Short Proof of an Entropy Upper Bound for Transport Plans

Let $r, c \in \Sigma_d$, and let $P \in \mathcal{U}(r, c)$, where

$$\mathcal{U}(r, c) = \left\{ P \in \mathbb{R}_+^{d \times d} : P\mathbf{1} = r, \quad P^\top \mathbf{1} = c \right\}.$$

In other words,

$$\sum_j P_{ij} = r_i, \quad \sum_i P_{ij} = c_j.$$

We define the Kullback–Leibler divergence between P and $rc^\top$ by

$$\text{KL}(P \| rc^\top) = \sum_{i,j} P_{ij} \log \left(\frac{P_{ij}}{r_i c_j} \right).$$

Assume that all components of r and c are strictly positive. Then

$$\text{KL}(P \| rc^\top) = \sum_{i,j} P_{ij} \log \left(\frac{P_{ij}}{r_i c_j} \right).$$

Hence,

$$\text{KL}(P \| rc^\top) = \sum_{i,j} P_{ij} \log P_{ij} - \sum_{i,j} P_{ij} \log r_i - \sum_{i,j} P_{ij} \log c_j.$$

Let

$$h(P) = - \sum_{i,j} P_{ij} \log P_{ij}, \quad h(r) = - \sum_i r_i \log r_i, \quad h(c) = - \sum_j c_j \log c_j.$$

Then

$$\begin{aligned} \sum_{i,j} P_{ij} \log r_i &= \sum_i \left(\sum_j P_{ij} \right) \log r_i \\ &= \sum_i r_i \log r_i \\ &= -h(r), \end{aligned}$$

and similarly,

$$\begin{aligned} \sum_{i,j} P_{ij} \log c_j &= \sum_j \left(\sum_i P_{ij} \right) \log c_j \\ &= \sum_j c_j \log c_j \\ &= -h(c). \end{aligned}$$

Therefore,

$$\text{KL}(P \| rc^\top) = h(r) + h(c) - h(P).$$

We now show that this quantity is nonnegative. Recall that, for all $x > 0$,

$$\log x \leq x - 1.$$

Equivalently,

$$-\log x \geq 1 - x.$$

Applying this inequality with

$$x = \frac{r_i c_j}{P_{ij}},$$

we obtain

$$\log \left(\frac{P_{ij}}{r_i c_j} \right) = -\log \left(\frac{r_i c_j}{P_{ij}} \right) \geq 1 - \frac{r_i c_j}{P_{ij}}.$$

Thus,

$$\begin{aligned} \text{KL}(P \| rc^\top) &= \sum_{i,j} P_{ij} \log \left(\frac{P_{ij}}{r_i c_j} \right) \\ &\geq \sum_{i,j} P_{ij} \left(1 - \frac{r_i c_j}{P_{ij}} \right) \\ &= \sum_{i,j} P_{ij} - \sum_{i,j} r_i c_j \\ &= 1 - 1 \\ &= 0. \end{aligned}$$

Hence,

$$\text{KL}(P \| rc^\top) \geq 0.$$

Since

$$\text{KL}(P \| rc^\top) = h(r) + h(c) - h(P),$$

we finally obtain

$$h(P) \leq h(r) + h(c).$$